

1 Three Main Connectivity Modes in Power BI

Power BI provides **three main data connectivity modes** for connecting to data sources:

Connectivity Mode	Description	When to Use
Import Mode	Imports a copy of the data into Power BI. The data is stored in the Power BI dataset.	When you want fast performance and periodic refresh (not real-time).
DirectQuery Mode	Does not import data — queries run directly on the source every time you interact with the report.	When you need real-time or near real-time data updates.
Live Connection	Directly connects to Analysis Services (SSAS, Azure Analysis Services). The model and calculations stay on the server.	When the data model is already built in Analysis Services and should not be duplicated.

2 Two Types of Gateways in Power BI

Power BI gateways act as a **bridge** between your **on-premises data sources** and the **Power BI Service (cloud)**.

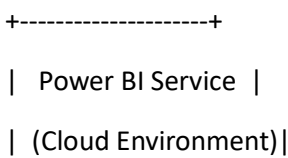
Type of Gateway	Description	Usage Scenario
1. Personal Gateway	Designed for individual use. Runs under a single user account. Cannot be shared with others.	Best for personal reports or small-scale use.
2. Standard (Enterprise) Gateway	Used by multiple users; supports central data access and shared connections.	Best for organizational and enterprise-level environments.

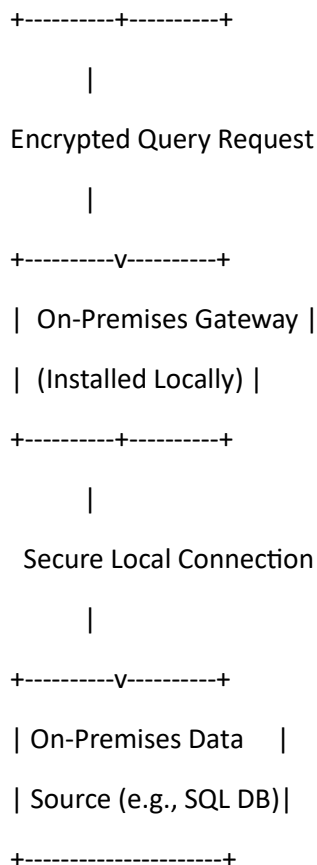
3 How Gateway Works (with Diagram Explanation)

Working Mechanism:

1. The **user publishes** a Power BI report from **Power BI Desktop** to the **Power BI Service (cloud)**.
2. The **Power BI Service** does not store on-premises data; it sends a **query request** to the **On-premises Data Gateway**.
3. The **Gateway** acts as a secure bridge — it **encrypts and transmits queries** between the cloud and the local data source.
4. The **Gateway** then connects to the **on-premises data source** (like SQL Server, Oracle, etc.), **executes the query**, and **returns the results** to Power BI Service.
5. The **Service updates dashboards/reports** with the latest data.

Diagram (Text Description)





Flow:
Power BI Service → Gateway → Data Source → Gateway → Power BI Service

4 Difference Between Cloud-Based Data and On-Premises Data

Aspect	Cloud-Based Data	On-Premises Data
Location	Stored on online/cloud servers (e.g., Azure, Salesforce, Google BigQuery).	Stored within the organization’s internal servers.
Access	Accessible via internet from anywhere.	Accessible only within the organization’s network or VPN.
Gateway Requirement	❌ No gateway needed (Power BI can connect directly).	✅ Requires a gateway for secure connection.
Data Refresh	Automatic or scheduled refresh through cloud connection.	Refresh occurs through on-premises gateway.
Example Sources	Azure SQL Database, SharePoint Online, Snowflake, Salesforce.	SQL Server, Oracle, SAP HANA, Excel on local drive.

5 Steps to Use Gateway in Power BI

Here’s the **step-by-step process** 🖱️

Step 1: Download the Gateway

- Go to [Power BI Gateway Download Page](#).

- Choose the **Standard mode** (recommended for most organizations).
-

Step 2: Install the Gateway

- Run the downloaded setup file.
 - Choose “**On-premises data gateway (recommended)**” during installation.
 - Sign in with your **Power BI account**.
-

Step 3: Configure the Gateway

- Register your gateway with your **organization’s Power BI tenant**.
 - Assign a name and **recovery key** (used for backup or migration).
-

Step 4: Add Data Source to Gateway

- In **Power BI Service** → **Settings** → **Manage Gateways** → **Add Data Source**.
 - Choose the **data source type** (SQL Server, Oracle, etc.).
 - Enter **server, database, and credentials**.
-

Step 5: Map Dataset to Gateway

- Go to your **Power BI dataset** in the Service.
 - Under **Settings** → **Gateway connection**, choose the gateway and map it to your dataset.
-

Step 6: Schedule Data Refresh (Optional)

- Enable **scheduled refresh** in Power BI Service.
 - The gateway will fetch data automatically from the local source at defined intervals.
-